EMCH 368 Mechatronics
Fall 2026
Practice Set 2 Solutions

100 points

Answer key. Each problem repeats the practice questions with completed answer boxes. A separate worked-solution page follows each problem. This set uses new scenarios and tasks: calibration equations, design inequalities, circuit analysis, signal filtering, and data calculations. It is a companion to Practice Set 1.

Homework connections

All references below point to `HW_fall_2026.pdf`. Page numbers are the PDF page positions, counted from 1; they are not the repeated Blackboard page numbers. Related homework questions cover the underlying concepts; these practice questions extend them.

Set 2	Topic	Related homework	PDF pp.
1	Calibration correction	HW03 Q2, Q5, Q10	13–16
2	Accuracy constraints	HW03 Q5, Q10	14–16
3	Encoder sizing	HW03 Q7, Q9	15–16
4	Light-triggered switch	HW04 Q1, Q8	17, 20–21
5	Amplifier design	HW04 Q2, Q5, Q10	17–19, 21–22
6	Sensor settling constraint	HW03 Q4; HW04 Q9	14, 21
7	Null-balance measurement	HW05 Q9	26–27
8	Filter selection	HW05 Q5, Q8	24–26
9	Multiplexed acquisition	HW06 Q4, Q8, Q10	29–32
10	Digital transmission	HW05 Q4; HW06 Q3, Q5, Q7	24, 29–30

Independent practice material based on the supplied homework. Question and solution formatting follows the supplied HW2 examples.

Problem 1. (10 pt) Build and test a calibration correction

Related homework: HW03 Q2, Q5, Q10 (PDF pp. 13–16): sensitivity and measurement error.

A pressure instrument is assumed to follow a straight-line model

$$I = mp + b,$$

where p is true pressure in kPa and I is the instrument's displayed value in kPa. A reference test gives $I = 12.0$ at $p = 0$ and $I = 392.0$ at $p = 400.0$ kPa.

(a) (3 pt) Determine m and b, and write the forward calibration equation.

(b) (3 pt) Rearrange the equation to correct any displayed value. Find the corrected pressure when the instrument displays 250.0 kPa.

(c) (4 pt) In an independent check at a true pressure of 300.0 kPa, the display reads 299.0 kPa. Calculate the uncorrected signed error and the signed error after applying your correction. Give the corrected error as a percentage of the 400.0 kPa span. Use indicated minus true for signed error.

(a) Slope, offset, and forward equation
$m = 0.950; \quad b = 12.0\ \text{kPa}$ $I = 0.950p + 12.0$, with I, p in kPa.

(b) Correction equation and corrected pressure
$p_{\text{corr}} = (I - 12.0)/0.950$ At $I = 250.0$: $p_{\text{corr}} = 250.526\ \text{kPa}$.

(c) Check-point errors
Uncorrected error: -1.00 kPa. Corrected pressure: 302.105 kPa. Corrected error: $+2.105$ kPa, or $+0.5263\%$ of span.

Problem 1. Worked solution

(a) The slope and offset follow from the two reference points:

$$m = \frac{392.0 - 12.0}{400.0 - 0} = 0.950, \qquad b = 12.0\ \text{kPa}.$$

Thus $I = 0.950p + 12.0$. The slope is dimensionless because both quantities are pressures in the same units.

(b) Subtract the offset, then divide by the slope:

$$p_{\text{corr}} = \frac{I - 12.0}{0.950}.$$

For a display of 250.0 kPa,

$$p_{\text{corr}} = \frac{250.0 - 12.0}{0.950} = 250.526316\ \text{kPa}.$$

(c) The uncorrected signed error is $299.0 - 300.0 = -1.00$ kPa. Applying the two-point correction to this independent reading gives

$$p_{\text{corr}} = \frac{299.0 - 12.0}{0.950} = 302.105263\ \text{kPa}.$$

The corrected signed error is therefore

$$e = 302.105263 - 300.0 = +2.105263\ \text{kPa},$$

$$e_{\%\text{span}} = 100\frac{2.105263}{400.0} = +0.526316\%.$$

The independent point does not lie exactly on the assumed straight line. Two-point calibration removes the fitted offset and scale error, but it does not guarantee zero error at every other point.

Problem 2. (10 pt) Compare two accuracy specifications

Related homework: HW03 Q5, Q10 (PDF pp. 14–16): full-span and signed measurement error.

Two pressure sensors cover 0–600 kPa. Their maximum absolute-error specifications are:

- Sensor A: $\pm 0.25\%$ of the 600 kPa span.
- Sensor B: $\pm(0.40\%$ of reading $+\ 0.200\ \mathrm{kPa})$.

For comparing the specifications, use the nominal positive pressure p as the reading in Sensor B's error formula. Add its two specified contributions directly.

(a) (3 pt) Express each maximum error magnitude E_A and $E_B(p)$ in kPa. Find the pressure where their error bounds are equal.

(b) (3 pt) Compute both error bounds at 75.0 kPa and at 500.0 kPa. Select the sensor with the smaller bound at each pressure.

(c) (4 pt) A new requirement limits maximum error to 0.500% of the nominal pressure. Find the lowest pressure where A qualifies and the lowest pressure where B qualifies. Give each qualifying interval within 0–600 kPa.

(a) Error functions and crossover
$E_A = 1.50\ \mathrm{kPa}; \quad E_B = 0.00400p + 0.200\ \mathrm{kPa}$ Equal bounds at $p = 325\ \mathrm{kPa}$.

(b) Comparisons at the two pressures
At 75.0 kPa: A $= 1.50$, B $= 0.500$ kPa; choose B. At 500.0 kPa: A $= 1.50$, B $= 2.20$ kPa; choose A.

(c) Minimum pressures and qualifying intervals
A: minimum 300 kPa; qualifies on $[300, 600]$ kPa. B: minimum 200 kPa; qualifies on $[200, 600]$ kPa.

Problem 2. Worked solution

(a) Convert percentages to fractions:

$$E_A = 0.00250(600) = 1.50 \text{ kPa},$$

$$E_B(p) = 0.00400p + 0.200 \text{ kPa},$$

where p is entered in kPa. Set the two bounds equal:

$$1.50 = 0.00400p + 0.200 \quad \Rightarrow \quad p = 325 \text{ kPa}.$$

B has the smaller bound below 325 kPa; A has the smaller bound above 325 kPa.

(b) At 75.0 kPa, $E_B = 0.00400(75.0) + 0.200 = 0.500$ kPa, less than A's 1.50 kPa. At 500.0 kPa, $E_B = 0.00400(500.0) + 0.200 = 2.20$ kPa, greater than A's 1.50 kPa.

(c) For positive pressure, the allowable absolute error is $0.00500p$. For A,

$$1.50 \le 0.00500p \quad \Rightarrow \quad p \ge 300 \text{ kPa}.$$

For B,

$$0.00400p + 0.200 \le 0.00500p \quad \Rightarrow \quad p \ge 200 \text{ kPa}.$$

Intersect these results with the specified 0–600 kPa operating range. The resulting intervals are $[300, 600]$ kPa for A and $[200, 600]$ kPa for B. At zero pressure, neither nonzero absolute-error bound meets a zero allowable error.

Problem 3. (10 pt) Specify an encoder before buying it

Related homework: HW03 Q7, Q9 (PDF pp. 15–16).

A tracking wheel has radius 25.0 mm. Its encoder produces one counted pulse per slot per revolution. The linear distance between consecutive pulses must be no greater than 0.200 mm. Assume no slip and do not use quadrature multiplication.

(a) (4 pt) Derive the minimum slot count and round it in the direction needed to satisfy the specification.

(b) (3 pt) Available encoders have 720, 800, or 1024 slots. Choose the lowest-count model that meets the requirement and calculate its linear distance per pulse.

(c) (3 pt) Keeping your chosen encoder, what is the largest wheel radius that still meets the 0.200 mm requirement? Would a replacement wheel of radius 30.0 mm pass?

(a) Design inequality and minimum integer slot count
$\frac{2\pi r}{N} \leq 0.200 \text{ mm}$ $N \geq 785.398 \quad \Rightarrow \quad N_{\min} = 786$ slots.

(b) Selected encoder and distance per pulse
Choose 800 slots. $\Delta x = 0.196350 \text{ mm/pulse}$.

(c) Maximum radius and replacement decision
$r_{\max} = 25.4648 \text{ mm}$. The 30.0 mm wheel fails: $\Delta x = 0.235619$ mm/pulse.

Problem 3. Worked solution

One revolution moves the wheel by its circumference, $2\pi r$. With N pulses per revolution, the displacement increment is

$$\Delta x = \frac{2\pi r}{N}.$$

(a) Requiring $\Delta x \le 0.200$ mm gives

$$N \ge \frac{2\pi(25.0)}{0.200} = 785.398.$$

The number of slots is an integer. Round *up* to 786, because rounding down would make the displacement increment too large.

(b) The 720-slot model fails because $2\pi(25)/720 = 0.218166$ mm/pulse. The 800-slot model gives

$$\Delta x = \frac{2\pi(25.0)}{800} = 0.196350 \ \mathrm{mm/pulse},$$

so it meets the limit. The 1024-slot model also passes but is not the lowest-count acceptable model.

(c) Solve the same design constraint for radius:

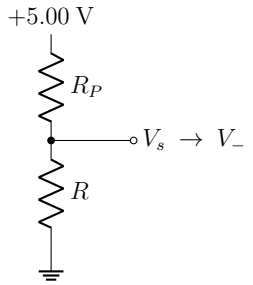
$$r \le \frac{N\Delta x_{\max}}{2\pi} = \frac{800(0.200)}{2\pi} = 25.4648 \ \mathrm{mm}.$$

At $r = 30.0$ mm, $\Delta x = 2\pi(30)/800 = 0.235619$ mm/pulse, which exceeds the specification.

Problem 4. (10 pt) Design a light-triggered switch

Related homework: HW04 Q1, Q8 (PDF pp. 17, 20–21).

A photoresistor R_P is connected above a fixed resistor R in a 5.00 V divider. Its midpoint is V_s. The photoresistor is $2.00\ \text{k}\Omega$ in bright light, $30.0\ \text{k}\Omega$ at the desired switching light level, and $100\ \text{k}\Omega$ in darkness. A comparator has $V_+ = 2.00$ V and $V_- = V_s$. Model its output levels as 0 and 5.00 V. Input currents are zero.



(a) (4 pt) Select R so that $V_s = 2.00$ V at the desired switching light level.

(b) (4 pt) Determine V_s and the comparator output in bright light and in darkness.

(c) (2 pt) A lamp driver turns on when the comparator output is high. Solve an inequality for the range of R_P that turns the lamp on. Exclude the exact switching threshold.

(a) Fixed resistor
$R = 20.0\ \text{k}\Omega$

(b) Bright and dark conditions
Bright: $V_s = 4.545$ V; output $= 0$ V. Dark: $V_s = 0.8333$ V; output $= 5.00$ V.

(c) Resistance range for lamp activation
$R_P > 30.0\ \text{k}\Omega$ gives a high output and turns the lamp on.

Problem 4. Worked solution

The divider output is

$$V_s = 5\frac{R}{R_P + R}.$$

(a) Use $R_P = 30.0\ \text{k}\Omega$ and $V_s = 2.00$ V. Expressing resistances in kΩ,

$$2 = \frac{5R}{30 + R} \quad \Rightarrow \quad 60 + 2R = 5R \quad \Rightarrow \quad R = 20.0\ \text{k}\Omega.$$

(b) In bright light,

$$V_s = 5\frac{20}{2 + 20} = 4.54545\ \text{V}.$$

Since $V_+ - V_- = 2.00 - 4.54545 < 0$, the comparator output is 0 V. In darkness,

$$V_s = 5\frac{20}{100 + 20} = 0.833333\ \text{V}.$$

Now $V_+ - V_- = 2.00 - 0.833333 > 0$, so the output is 5.00 V.

(c) A high output requires $2.00 > V_s$. With resistances in kΩ,

$$2 > \frac{5(20)}{R_P + 20} \quad \Rightarrow \quad 2R_P + 40 > 100 \quad \Rightarrow \quad R_P > 30.0\ \text{k}\Omega.$$

At equality, this ideal comparator model does not determine a unique output.

Problem 5. (10 pt) Design an amplifier and check its limits

Related homework: HW04 Q2, Q5, Q10 (PDF pp. 17–19, 21–22).

A sensor is first connected to an ideal voltage follower, then to an inverting amplifier. The desired mapping is $V_{\text{in}} = -0.300$ V to $V_o = +1.80$ V and $V_{\text{in}} = +0.900$ V to $V_o = -5.40$ V. The inverting stage has $R_i = 12.0\ \text{k}\Omega$. Both op-amps have output limits of ± 4.50 V. Assume input common-mode limits are not a concern.

(a) (3 pt) Determine the required linear gain and feedback resistor R_f.

(b) (3 pt) Can the full requested input range be processed without clipping? Find the largest positive sensor input that stays within the output limits, and the actual output at $+0.900$ V.

(c) (2 pt) For a separate loading test, model the sensor as a 0.450 V source in series with $6.00\ \text{k}\Omega$. Calculate the final amplifier output with the follower present and with the follower removed (the sensor then drives R_i directly).

(d) (2 pt) At $V_{\text{in}} = +0.900$ V with the follower present, calculate the inverting-input node voltage using the actual clipped output. The non-inverting input is grounded.

(a) Gain and feedback resistor
$G = -6.00; \qquad R_f = 72.0\ \text{k}\Omega$

(b) Feasibility, input limit, and clipped output
Full range: not feasible with these output limits. Largest positive linear input: $+0.750$ V. At $+0.900$ V, actual output: -4.50 V.

(c) Final output with and without the follower
With follower: $V_o = -2.70$ V. Without follower: $V_o = -1.80$ V.

(d) Inverting-input node voltage under clipping
$V_- = +0.128571$ V; it is not equal to the grounded non-inverting input.

Problem 5. Worked solution

(a) The required slope is

$$G = \frac{-5.40 - 1.80}{0.900 - (-0.300)} = -6.00.$$

The mapping passes through the origin, so no offset is needed. An inverting amplifier has $G = -R_f/R_i$, giving

$$R_f = 6R_i = 6(12.0) = 72.0\ \text{k}\Omega.$$

(b) Linear operation requires

$$|-6V_{\text{in}}| \le 4.50 \quad \Rightarrow \quad |V_{\text{in}}| \le 0.750\ \text{V}.$$

Thus the requested interval is linear only from -0.300 to $+0.750$ V. The largest positive allowed input is $+0.750$ V at the ideal boundary. At $+0.900$ V, the desired output is -5.40 V, but the output clips to -4.50 V.

(c) The ideal follower draws no sensor current, so it reproduces the open-circuit sensor voltage, 0.450 V. The final output is $-6(0.450) = -2.70$ V. Without the follower, the sensor resistance adds to the input resistance. With the inverting node at virtual ground,

$$I = \frac{0.450}{(6.00 + 12.0)\,10^3} = 25.0\ \mu\text{A},$$

$$V_o = -IR_f = -(25.0 \times 10^{-6})(72.0 \times 10^3) = -1.80\ \text{V}.$$

Both results are within the output limits, so linear feedback is valid for this loading test.

(d) At input 0.900 V, use the actual output -4.50 V in node-current balance. The op-amp input draws no current:

$$\frac{V_- - 0.900}{12.0} + \frac{V_- - (-4.50)}{72.0} = 0,$$

where resistances are in kΩ. Thus

$$V_- = \frac{72(0.900) + 12(-4.50)}{72 + 12} = 0.128571\ \text{V}.$$

The non-inverting input remains at 0 V. Saturation prevents the output from maintaining the virtual-short condition.

Problem 6. (10 pt) Choose a sensor for a settling deadline

Related homework: HW03 Q4; HW04 Q9 (PDF pp. 14, 21).

Two first-order temperature sensors are initially at $90.0\,^\circ\mathrm{C}$. Each is moved into a bath held at $30.0\,^\circ\mathrm{C}$. Sensor A has time constant 8.00 s; sensor B has time constant 20.0 s. Ignore dead time, noise, and steady-state error.

The acceptance requirement is: at 30.0 s after immersion, the reading must be within $2.00\,^\circ\mathrm{C}$ of the bath temperature.

(a) (4 pt) Calculate the remaining temperature error at 30.0 s for each sensor, and decide which sensors pass.

(b) (4 pt) Derive the largest time constant a first-order sensor could have and still pass this requirement.

(c) (2 pt) If sensor B must be used, calculate the earliest time at which its error is no greater than $2.00\,^\circ\mathrm{C}$.

(a) Remaining errors and pass/fail
A: $1.411\,^\circ\mathrm{C}$ above the bath; passes. B: $13.39\,^\circ\mathrm{C}$ above the bath; fails.

(b) Maximum permissible time constant
$\tau \le \dfrac{30.0}{\ln(60.0/2.00)} = 8.82042\ \mathrm{s}$

(c) Earliest acceptable measurement time for B
$t_{\min} = 20.0 \ln 30 = 68.0239\ \mathrm{s}.$

Problem 6. Worked solution

For cooling from $T_0 = 90$ to $T_\infty = 30\,^\circ\mathrm{C}$,

$$T(t) = 30 + 60e^{-t/\tau}.$$

The remaining error magnitude is

$$E(t) = |T(t) - 30| = 60e^{-t/\tau}.$$

(a) At $t = 30.0$ s,

$$E_A = 60e^{-30/8} = 1.41106\,^\circ\mathrm{C} \le 2.00\,^\circ\mathrm{C},$$

$$E_B = 60e^{-30/20} = 13.3878\,^\circ\mathrm{C} > 2.00\,^\circ\mathrm{C}.$$

Sensor A passes; sensor B fails. Their readings are 31.4111 and $43.3878\,^\circ\mathrm{C}$, respectively.

(b) Apply the allowed-error requirement directly:

$$60e^{-30/\tau} \le 2 \quad \Rightarrow \quad \frac{30}{\tau} \ge \ln 30.$$

Because $\tau > 0$,

$$\boxed{\tau \le \frac{30}{\ln 30} = 8.82042 \text{ s}.}$$

A larger time constant leaves too much of the original temperature difference at the deadline.

(c) Set sensor B's remaining error equal to the allowed limit:

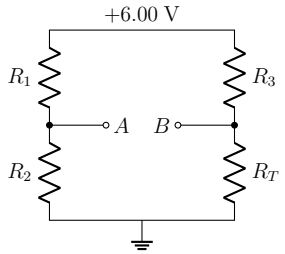
$$60e^{-t/20} = 2 \quad \Rightarrow \quad t = 20 \ln 30 = 68.0239 \text{ s}.$$

The error decreases monotonically, so B meets the requirement for $t \ge 68.0239$ s.

Problem 7. (10 pt) Use a bridge as a null-measurement instrument

Related homework: HW05 Q9 (PDF pp. 26–27); resistor positions are defined below.

A resistance thermometer R_T is measured with a bridge. The left branch has $R_1 = 2.00\ \text{k}\Omega$ above node A and $R_2 = 3.00\ \text{k}\Omega$ below A. The right branch has adjustable R_3 above node B and R_T below B. The supply is 6.00 V, and the detector draws no current.



(a) (4 pt) The detector reads zero when $R_3 = 1.60\ \text{k}\Omega$. Determine R_T and the voltage of each midpoint relative to ground.

(b) (3 pt) For this thermometer, resistance varies linearly from $3.00\ \text{k}\Omega$ at $20.0\,^\circ\text{C}$ to $1.80\ \text{k}\Omega$ at $60.0\,^\circ\text{C}$. Find the measured temperature.

(c) (3 pt) The thermometer later reaches $60.0\,^\circ\text{C}$. What value of R_3 restores the null? With that setting and a reduced supply of 3.00 V, calculate both midpoint voltages and $V_B - V_A$.

(a) Unknown resistance and midpoint voltages
$R_T = 2.40\ \text{k}\Omega; \quad V_A = V_B = 3.60\ \text{V}$

(b) Measured temperature
$T = 40.0\,^\circ\text{C}$

(c) New adjustment and midpoint voltages at 3.00 V
$R_3 = 1.20\ \text{k}\Omega$. $V_A = V_B = 1.80$ V; $V_B - V_A = 0$ V.

Problem 7. Worked solution

(a) At a null, the midpoint voltages are equal:

$$6\frac{R_2}{R_1+R_2} = 6\frac{R_T}{R_3+R_T}.$$

Cross-multiplication gives $R_2R_3 = R_1R_T$, so

$$R_T = \frac{R_2R_3}{R_1} = \frac{3.00(1.60)}{2.00} = 2.40\ \text{k}\Omega.$$

Each midpoint is at

$$V_A = 6\frac{3}{2+3} = 3.60\ \text{V}, \qquad V_B = 6\frac{2.4}{1.6+2.4} = 3.60\ \text{V}.$$

(b) Use the calibration supplied in the problem, rather than a generic temperature coefficient. Linear interpolation gives

$$T = 20.0 + \frac{2.40-3.00}{1.80-3.00}(60.0-20.0) = 40.0\,^\circ\text{C}.$$

(c) At $60.0\,^\circ\text{C}$, $R_T = 1.80\ \text{k}\Omega$. Restore balance by setting

$$R_3 = \frac{R_1R_T}{R_2} = \frac{2.00(1.80)}{3.00} = 1.20\ \text{k}\Omega.$$

The supply voltage cancels from the balance equation. An ideal change from 6 to 3 V therefore leaves the resistance setting unchanged, although the balanced midpoint voltages decrease from 3.60 to 1.80 V.

Problem 8. (10 pt) Remove two different kinds of interference

Related homework: HW05 Q5, Q8 (PDF pp. 24–26).

A measurement contains useful components at 40 Hz and 200 Hz, unwanted interference at 120 Hz, and unwanted noise at 3000 Hz:

$$x(t) = 2\sin(2\pi 40t) + 0.8\sin(2\pi 120t) \\ + \sin(2\pi 200t) + 0.3\sin(2\pi 3000t) \quad [\mathrm{V}].$$

Consider the following ideal filters. Each has gain 1 in its passband and gain 0 in its stopband, with zero phase shift for this exercise.

A. Low-pass: passes frequencies below 1000 Hz.

B. High-pass: passes frequencies above 150 Hz.

C. Band-stop: rejects 100–140 Hz and passes other frequencies.

D. Band-pass: passes 100–250 Hz only.

(a) (4 pt) Choose two filters to cascade so that both useful components remain and both unwanted components are removed. Explain the purpose of each filter.

(b) (3 pt) Write the resulting output $y(t)$ under the ideal assumptions.

(c) (3 pt) Write the output signal when only filter B is used, then when only filter D is used.

(a) Selected filters and roles
A and C. The low-pass removes 3000 Hz; the band-stop removes 120 Hz while retaining 40 and 200 Hz.

(b) Output signal
$y(t) = 2\sin(2\pi 40t) + \sin(2\pi 200t) \quad [\mathrm{V}]$

(c) Outputs for B alone and D alone
$y_B = \sin(2\pi 200t) + 0.3\sin(2\pi 3000t)$ V. $y_D = 0.8\sin(2\pi 120t) + \sin(2\pi 200t)$ V.

Problem 8. Worked solution

(a) A low-pass filter alone preserves both useful frequencies but cannot remove an interference frequency lying between them. The band-stop filter supplies that missing rejection.

The cascade's gains multiply at each frequency:

Frequency (Hz)	A gain	C gain	Combined gain
40	1	1	1
120	1	0	0
200	1	1	1
3000	0	1	0

Under the stated ideal assumptions, either cascade order gives the same result.

(b) The surviving components keep their amplitudes and phases:

$$y(t) = 2\sin(2\pi 40t) + \sin(2\pi 200t) \quad [\mathrm{V}].$$

(c) Filter B retains only 200 and 3000 Hz:

$$y_B(t) = \sin(2\pi 200t) + 0.3\sin(2\pi 3000t) \quad [\mathrm{V}].$$

Filter D retains only 120 and 200 Hz:

$$y_D(t) = 0.8\sin(2\pi 120t) + \sin(2\pi 200t) \quad [\mathrm{V}].$$

Both lose the desired 40 Hz component and leave an unwanted component.

Problem 9. (10 pt) Diagnose a multiplexed data-acquisition system

Related homework: HW06 Q4, Q8, Q10 (PDF pp. 29–32).

A six-channel DAQ shares one ADC through a multiplexer. It samples channels 1 through 6 repeatedly, with one conversion per channel per scan. The ADC performs 6000 conversions/s in total. Assume equally spaced conversions and negligible switching overhead. Each channel contains desired frequencies up to 650 Hz.

(a) (3 pt) Find the sample rate of one channel and the time between successive samples of that same channel.

(b) (3 pt) A 650 Hz sinusoid appears as a lower-frequency sinusoid in the recorded data. Find its alias frequency between 0 and half the per-channel sample rate.

(c) (2 pt) To preserve the entire 0–650 Hz band, determine the strict lower bound on total conversion rate. Which available setting, 7200 or 12000 conversions/s, is sufficient under the ideal sampling criterion?

(d) (2 pt) At 12000 total conversions/s, what is the largest integer number of equally sampled channels that can each preserve a 0–650 Hz band under the same strict sampling criterion?

(a) Per-channel rate and sample interval
$f_{s,\mathrm{ch}} = 1000\ \mathrm{samples/s}; \quad T_{s,\mathrm{ch}} = 1.00\ \mathrm{ms}$

(b) Recorded alias frequency
$f_{\mathrm{alias}} = 350\ \mathrm{Hz}$

(c) Total-rate requirement and selected setting
$F_{\mathrm{total}} > 7800\ \mathrm{conversions/s}$. Choose 12000 conversions/s.

(d) Maximum channel count at 12000 conversions/s
$M_{\max} = 9$ channels.

Problem 9. Worked solution

(a) The aggregate conversion rate is divided among six channels:

$$f_{s,\mathrm{ch}} = \frac{6000}{6} = 1000 \text{ samples/s}, \qquad T_{s,\mathrm{ch}} = \frac{1}{1000} = 1.00 \text{ ms}.$$

The time between *adjacent ADC conversions* is $1/6000$ s, but those conversions belong to different channels.

(b) The per-channel Nyquist frequency is $1000/2 = 500$ Hz. Folding the 650 Hz component into the 0–500 Hz interval gives

$$f_{\mathrm{alias}} = |650 - 1000| = 350 \text{ Hz}.$$

For sampled sinusoids, the aliased representation may also involve a phase change; the frequency observed is 350 Hz.

(c) Require a per-channel rate strictly above twice the desired highest frequency:

$$\frac{F_{\mathrm{total}}}{6} > 2(650) \quad \Rightarrow \quad F_{\mathrm{total}} > 7800 \text{ conversions/s}.$$

The 7200 setting gives only 1200 samples/s per channel and fails. The 12000 setting gives 2000 samples/s per channel and passes the ideal criterion.

(d) For M channels,

$$\frac{12000}{M} > 2(650) \quad \Rightarrow \quad M < \frac{12000}{1300} = 9.23077.$$

The largest permitted integer is 9. Nine channels receive 1333.33 samples/s each and pass; ten receive only 1200 samples/s each and fail.

Problem 10. (10 pt) Decode and verify received data

Related homework: HW05 Q4; HW06 Q3, Q5, Q7 (PDF pp. 24, 29–30).

An unsigned 8-bit measurement word is transmitted with one separate even-parity bit. The data word is `01101010`. The parity bit is excluded from the numerical value.

(a) (3 pt) Decode the original data word into decimal and determine its even-parity bit.

(b) (2 pt) A receiver obtains data `01101110` with the original parity bit unchanged. Does the parity check detect an error? Show the count of 1s.

(c) (3 pt) Another transmission arrives as `01001011`, again with the original parity bit unchanged. Does this pass parity? Decode its decimal value and explain what this demonstrates.

(d) (2 pt) The ADC assigns code k to voltages in $[kq, (k+1)q)$, with $q = 0.0200$ V. It reports the midpoint $(k+0.5)q$ as its voltage estimate. For the original word, calculate the input-voltage interval, midpoint estimate, and maximum quantization-error magnitude.

(a) Original decimal value and parity bit
$01101010_2 = 106_{10}$; $\quad p = 0$ (four data 1s).

(b) First received word
Detected: five data 1s plus parity 0 gives an odd total, so the even-parity check fails.

(c) Second received word
Passes parity: four data 1s plus parity 0 is even. Received value $= 75$. Two flipped data bits can escape a single parity check.

(d) Voltage interval, estimate, and error magnitude
$V_{\text{in}} \in [2.120, 2.140)$ V; estimate $= 2.130$ V. Maximum error magnitude $= 0.0100$ V.

Problem 10. Worked solution

(a) Use binary place values $128, 64, 32, 16, 8, 4, 2, 1$:

$$01101010_2 = 64 + 32 + 8 + 2 = 106.$$

There are four 1s. The total is already even, so the appended parity bit is 0.

(b) The word `01101110` has five 1s. Adding the unchanged parity bit 0 leaves five 1s, so the receiver detects a parity mismatch. A parity check does not identify which bit is wrong.

(c) The word `01001011` has four 1s, so it passes with parity 0. Its value is

$$01001011_2 = 64 + 8 + 2 + 1 = 75.$$

Relative to the original, the 32's-place bit changed from 1 to 0 and the 1's-place bit changed from 0 to 1. Thus $106 - 32 + 1 = 75$. An even number of bit flips can preserve parity even though the data is wrong. Passing a parity check does not prove a word is correct.

(d) The original code is $k = 106$. Its lower and upper voltage limits are

$$V_{\text{low}} = 106(0.0200) = 2.120 \text{ V}, \qquad V_{\text{high}} = 107(0.0200) = 2.140 \text{ V}.$$

The upper bound is excluded under the specified code convention. The midpoint estimate is

$$\widehat{V} = (106 + 0.5)(0.0200) = 2.130 \text{ V}.$$

The maximum error magnitude is half a bin width, $q/2 = 0.0100$ V.